Towards In-the-wild 3D Plane Reconstruction from a Single Image

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Abstract

Single-image 3D plane reconstruction methods trained on a single domain fail to generalize to diverse in-the-wild scenes. Two key obstacles exist: (1) the absence of a large-scale mixed-domain benchmark with dense, high-resolution plane annotations enabling unified training and zero-shot evaluation, and (2) the lack of a method that handles cross-dataset geometric variability. We introduce the ***ZeroPlane Planar Benchmark****, spanning 14+ datasets with* $\sim$*560K high-resolution (*640×480*) plane annotations across indoor and outdoor scenes, and the* ***ZeroPlane*** *framework, a Transformer detector that decouples plane normal and offset prediction via exemplar-guided classification-then-regression and fuses auxiliary pixel-level geometry into plane queries. Mixed-domain training with ZeroPlane achieves state-of-the-art zero-shot performance on four held-out benchmarks: on NYUv2, depth recall@0.1m improves from 14.29% to 21.20% and normal recall@10° from 40.96% to 47.76%; on outdoor ParallelDomain, recall@3m rises from 35.49% to 46.26% and normal recall@10° from 35.11% to 66.13%.*

1. Introduction

Piecewise planar reconstruction from a single image is fundamental to 3D scene understanding, enabling compact, structured representations for robotics, augmented reality, and autonomous driving [7, 12, 28]. Despite significant progress [18, 19, 25, 30, 37], existing methods are predominantly trained and evaluated on single domains—most often indoor ScanNet [9] data—and degrade sharply when applied out-of-distribution. The challenge is compounded by the absence of a suitable cross-domain benchmark: without a unified dataset spanning multiple environments, neither cross-domain training nor fair zero-shot evaluation is possible. In contrast, the monocular depth estimation community has long demonstrated that mixing diverse training datasets yields robust zero-shot models [22].

In this work, we address both obstacles simultaneously. We introduce the **ZeroPlane Planar Benchmark**, a large-scale mixed-domain collection spanning 14+ datasets with $\sim$560K high-resolution (640×480) plane annotations generated via a unified plane-fitting pipeline. Zero-shot evaluation is defined on four held-out benchmarks: NYUv2 [27] and 7-Scenes [26] (indoor), and ParallelDomain [13, 21] and ApolloScape [16] (outdoor).

We further present **ZeroPlane**, a Transformer-based detector with DINOv2 [20] encoder and DPT-style [23] multi-scale pixel decoder. Its key contributions are: (i) *decoupled exemplar-guided prediction*—plane normals and offsets are each predicted by first classifying into K-means exemplar clusters and then regressing a residual, providing a shared geometric vocabulary across depth scales; and (ii) *geometry-enhanced plane queries*—auxiliary pixel depth and normal heads supply geometric priors that are fused into plane queries via attention.

Contributions:

- The ZeroPlane Planar Benchmark: $\sim$560K annotations across 14+ datasets enabling cross-domain training and zero-shot evaluation.
- ZeroPlane: a unified Transformer with decoupled exemplar-guided normal/offset prediction and geometry-enhanced query embeddings.
- Consistent state-of-the-art zero-shot generalization across four unseen benchmarks.

2. Related Work

Plane detection and reconstruction. PlaneNet [18] pioneered end-to-end plane reconstruction from a single RGB image. PlaneRCNN [19] extended Mask R-CNN [14] for simultaneous plane detection and parameter regression. PlaneAE [37] used associative embedding for arbitrary-count plane detection. PlaneTR [30] and PlaneRecTR [25] introduced Transformer-based frameworks with query learning for unified 3D plane recovery. PlaneFormers [1] addressed multi-view settings and PlanarRecon [34] tackled real-time video plane detection. All prior methods are trained and evaluated on single domains, motivating our cross-domain benchmark.

Planar benchmarks and generalization. Existing plane reconstruction benchmarks are confined to single-domain settings, with ScanNet [9] serving as the de facto standard for indoor evaluation and outdoor plane datasets remaining scarce. This single-domain focus prevents systematic study

of cross-domain generalization and hinders robust in-the-wild deployment. Our benchmark directly addresses this gap by unifying 14+ datasets under a consistent annotation protocol, enabling the first large-scale cross-domain plane reconstruction study.

Query-based detection and segmentation. DETR [5] introduced bipartite matching for set-prediction detection. Mask2Former [8] extended this with masked attention for universal image segmentation, serving as the foundation of our decoder.

Cross-domain depth and surface estimation. MiDaS [22] demonstrated that dataset mixing enables zero-shot depth transfer. Metric3D [15] presented a foundation model for zero-shot metric depth and surface normal estimation. DUSt3R [33] provides general 3D reconstruction via Transformer-based encoders. Omnidata [10] highlighted multi-task pretraining for generalization. AdaBins [2] introduced adaptive depth discretization, analogous to our exemplar-guided approach. DINOv2 [20] and DPT [23] provide strong visual backbones for dense prediction.

Large-scale pretraining for monocular scene understanding. DepthAnything [36] and Depth Pro [3] demonstrate that training monocular depth estimators on web-scale and large-scale labeled data yields strong zero-shot generalization to unseen scenes and sensors. SAM [17] showed analogous benefits for segmentation through large-scale interactive annotation. These works confirm that data scale and diversity are primary drivers of generalization—a principle our benchmark directly instantiates for structured 3D plane prediction, where both accurate segmentation and metric parameter estimation must jointly generalize across domains.

Zero-shot transfer and domain generalization. Achieving generalization to unseen domains without target-domain fine-tuning remains a central challenge in computer vision. Large-scale dataset mixing, as explored in MiDaS [22] and Omnidata [10], has proven effective for monocular depth estimation. Domain randomization and data augmentation strategies further improve robustness to distribution shift. Our work extends these insights to structured 3D plane prediction, showing that a combination of large-scale diverse training data and geometry-aware prediction heads yields robust zero-shot performance across both indoor and outdoor environments without any target-domain adaptation.

Table 1. Statistics of the datasets used in our work. Top: Datasets used for training and validation. Bottom: Datasets used for zero-shot evaluation.

Dataset	Indoor	Outdoor	Label	#Training Images	#Validation Images	#Planes
ScanNetv1 [9]	✓		RGB-D	47923	719	281268
ScanNetv2 [9]	✓		RGB-D	140773	859	1445183
Matterport3D [6]	✓		RGB-D	20854	943	227157
Replica [29]	✓		Synthetic	5533	400	90466
HM3D [35]	✓		Synthetic	79994	2000	1298635
DIODE (Indoor Split) [31]	✓		RGB-D	7350	731	37447
Taskonomy [38]	✓		RGB-D	123740	499	1450599
Synthia [24]		✓	Synthetic	24636	999	143698
Virtual KITTI [4, 11]		✓	Synthetic	17720	970	71171
Sanpo (Synthetic Split) [32]		✓	Synthetic	89854	488	647530
NYUv2 [27]	✓		RGB-D	-	654	4849
7-Scenes [26]	✓		RGB-D	-	758	5433
ApolloScape [16] (Stereo Split)		✓	Stereo	-	999	4241
ParallelDomain [13, 21]		✓	Synthetic	-	356	2350

3. The ZeroPlane Planar Benchmark

3.1. Overview and Data Sources

The ZeroPlane Planar Benchmark contains $\sim$560K plane annotations at 640×480, spanning 10 source datasets for training/validation and 4 held-out sets for zero-shot evaluation (Tab. 1).

Indoor RGB-D (real): ScanNetv1/v2 [9] provide large-scale annotated reconstructions (47,923 and 140,773 training images). Matterport3D [6] contributes building-scale panoramic views (20,854 images). DIODE (indoor) [31] adds high-resolution indoor RGB-D data and Taskonomy [38] provides large-scale multi-task annotations.

Indoor synthetic: Replica [29] and HM3D [35] offer photo-realistic indoor renderings with dense geometry.

Outdoor synthetic: Synthia [24], Virtual KITTI [4, 11], and Sanpo [32] cover diverse synthetic urban and pedestrian scenes.

Zero-shot evaluation: NYUv2 [27] and 7-Scenes [26] (indoor); ApolloScape (stereo) [16] and ParallelDomain [13, 21] (outdoor).

3.2. Annotation Pipeline

For each image, we back-project the available depth map to 3D and fit planes per panoptic segment using RANSAC. Where ground-truth semantics are available (e.g., ScanNet, HM3D), we use provided labels; otherwise, panoptic pseudo-labels are generated with Mask2Former [8]. We use an inlier distance threshold of ϵ=0.05 m; segments with fewer than 500 inlier points or an inlier ratio below 0.85 are discarded as non-planar. Across the full benchmark, annotations average 8.3 planes per image, with indoor scenes yielding denser annotations (avg. 10.2 planes) than outdoor scenes (avg. 5.7 planes), reflecting the more structured geometry of architectural environments. Domain-balanced sampling is applied during training to prevent high-volume datasets from dominating the loss. Annotations are dense binary plane masks with 3D parameters (n_x, n_y, n_z, d) at 640×480 resolution.

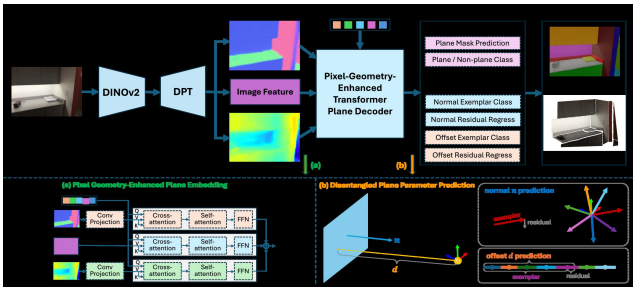

Figure 1. The **ZeroPlane** framework. A DINOv2 encoder and DPT pixel decoder produce multi-scale features fed to a Transformer decoder with plane queries. Decoupled exemplar-guided heads predict normals ($K_n = 7$) and offsets ($K_d = 20$). Geometry-enhanced embeddings fuse pixel depth/normal predictions into queries.

3.3. Evaluation Protocol

We evaluate plane detection, segmentation, and 3D parameter estimation. Segmentation uses RI↑, VOI↓, SC↑. Parameter accuracy uses plane recall under depth thresholds (indoor: 0.05/0.1/0.6 m; outdoor: 1/3/10 m) and normal thresholds (5°/10°/30°).

4. Methodology

4.1. Architecture Overview

ZeroPlane (Fig. 1) consists of: (1) an image encoder, (2) a DPT-style multi-scale pixel decoder, (3) a Transformer decoder with plane queries, (4) decoupled exemplar-guided normal and offset heads, and (5) auxiliary pixel depth/normal heads whose predictions are fused into queries via geometry-enhanced embeddings.

Encoder and pixel decoder. We use DINOv2 [20] (ViT-B/L) or DUSt3R [33] as the image encoder. A DPT-style [23] convolutional decoder assembles multi-scale features at $1/4$–$1/32$ resolution from the ViT stage outputs.

Transformer decoder. Following Mask2Former [8], N_q learnable plane queries are iteratively refined via masked cross-attention to image features and self-attention among queries. Per-query features drive mask prediction, plane class prediction, and geometry estimation.

4.2. Decoupled Exemplar-Guided Normal/Offset Prediction

Rather than directly regressing the 3D plane parameter $\mathbf{n}/d \in \mathbb{R}^3$, we factorize the prediction into independently estimated normals and offsets, each learned via classification-then-regression over exemplar clusters.

Normal exemplars. We pre-cluster K_n=7 normal exemplars $\{\hat{\mathbf{n}}_k\}$ via K-means over all training planes. For each query, the model predicts a softmax classification over clusters and a residual vector $\Delta\mathbf{n}$. The final normal $\mathbf{n} = \ell_2\text{-normalize}(\hat{\mathbf{n}}_k + \Delta\mathbf{n})$.

Offset exemplars. Similarly, K_d=20 offset exemplars are pre-clustered with depth split at 20 m to handle both indoor and outdoor scales. Classification selects a coarse depth bin; residual regression refines within the bin.

This factorization constrains predictions to a structured geometric manifold, reducing the effective search space. The shared exemplar vocabulary provides a common geometric language across datasets with heterogeneous depth scales, directly benefiting cross-domain generalization.

4.3. Geometry-Enhanced Plane Query Embeddings

Two auxiliary pixel-level heads predict dense depth and surface normal maps from the pixel decoder features. For each plane query, these predictions are aggregated within the predicted mask region and fused into the query embedding via cross-attention. The resulting geometry-enhanced embeddings provide explicit geometric priors for exemplar classification and residual regression, grounding each query's parameter estimate in the observed depth and normal structure. The auxiliary heads are jointly trained using ground-truth pixel maps, providing complementary supervision that strengthens feature representations for cross-domain transfer. Crucially, no target-domain data is required at inference: the geometry-enhanced embeddings rely entirely on the visual priors encoded by DINOv2, allowing the module to generalize to novel environments out of the box.

4.4. Training Objectives

We use Hungarian bipartite matching [5] to assign queries to ground-truth planes, minimizing classification, mask, and geometry costs. The training loss is:

$$\mathcal{L} = \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{mask}}\mathcal{L}_{\text{mask}} + \lambda_{\text{dice}}\mathcal{L}_{\text{dice}} + \lambda_{\text{n}}\mathcal{L}_{\text{normal}} + \lambda_{\text{d}}\mathcal{L}_{\text{offset}} + \lambda_{\text{pix}}\mathcal{L}_{\text{pixel}}, \tag{1}$$

where $\mathcal{L}_{\text{normal}}$ and $\mathcal{L}_{\text{offset}}$ each combine classification cross-entropy and residual ℓ_1 losses, and $\mathcal{L}_{\text{pixel}}$ covers auxiliary depth and normal losses with deep supervision applied across all decoder layers.

5. Experiments

5.1. Implementation Details

We train with the Detectron2 framework, batch size 16, AdamW optimizer (lr=10^{-4}, weight decay=0.05), at input resolution 640×480. DINOv2 weights are initialized from official pretrained checkpoints. Three variants are reported: DINO-B (ViT-B), DINO-L (ViT-L), and Dust3R (DUSt3R

encoder [33]). Training uses the 10 source datasets; zero-shot evaluation uses the four held-out test sets.

5.2. Zero-Shot Comparison

Tab. 2 reports zero-shot results on all four benchmarks. We compare against PlaneRCNN [19], PlaneAE [37], PlaneTR [30], and PlaneRecTR [25].

Single-domain (ScanNet) training. Even with single-domain training, our DINO-B backbone matches or exceeds all prior methods on NYUv2. It achieves RI=0.92, VOI=1.02, SC=0.75, and depth recall@0.1m of 11.03%, vs. PlaneRecTR's 10.13% at the same metrics. Normal recall@10° improves markedly from 34.42% to 39.88%.

Mixed-domain training. Mixed training yields substantially stronger zero-shot results. On **NYUv2**, ZeroPlane-DINO-B achieves recall@0.1m of 17.86% and normal@10° of 47.58%—up from 14.29% and 40.96% for PlaneRecTR (M). DINO-L improves recall@0.1m to 20.29% and normal@10° to 47.76%, while Dust3R achieves the best depth recall@0.1m of 21.20% and recall@0.6m of 56.03%. The segmentation quality (VOI) also improves from 1.01 to 0.93–0.97.

On **7-Scenes**, ZeroPlane-Dust3R achieves the best recall@0.1m of 19.14% (vs. 10.97%) and normal@10° of 36.11% (vs. 26.67%).

The gains are largest on **outdoor** benchmarks. On **ParallelDomain**, DINO-L achieves SC=0.87, VOI=0.57, recall@3m=44.72%, and normal@10°=66.13%, compared to SC=0.76, VOI=0.96, 35.49%, 35.11% for PlaneRecTR (M). Dust3R achieves the best recall@3m of 46.26%. On **ApolloScape**, DINO-L achieves recall@3m=46.40% and normal@10°=40.32%, vs. 43.06% and 33.32%.

5.3. Ablation Studies

Component contributions. Tab. 3 ablates the DPT pixel decoder, Cls-Reg heads, and Geo-Attn fusion on NYUv2. Under mixed training: adding DPT boosts recall@0.1m from 11.65% to 15.65% and normal@10° from 38.15% to 45.23%; adding Cls-Reg further improves recall@0.1m to 16.93% and normal@5° from 29.55% to 36.11%; adding Geo-Attn yields the best result: recall@0.1m=17.86% and normal@10°=47.58%. Each component also contributes under single-dataset training.

Training dataset composition. Tab. 4 studies the effect of dataset mixture. "Indoor Only" performs well on NYUv2 (recall@0.1m=18.70%) but collapses on ParallelDomain (1.19%). "Outdoor Only" reverses this pattern. The "Full Mixed" set (560K images) achieves the best overall performance, confirming that large-scale diverse training is critical for cross-domain generalization.

Training resolution. Tab. 5 shows that high-resolution training (640×480 vs. 192×256) significantly improves results when evaluated on high-resolution ground truth. On ParallelDomain, high-res training boosts recall@3m from 44.43% to 50.00% and normal@10° from 64.34% to 67.91%. On NYUv2, VOI improves from 0.93 to 0.84 and normal@10° from 47.87% to 50.34%.

5.4. Qualitative Results

Fig. 2 shows qualitative examples on all four zero-shot benchmarks. ZeroPlane correctly segments floors, walls, and furniture in indoor scenes, and road surfaces and building facades in outdoor scenes, demonstrating strong geometric understanding across domains. Fig. 3 illustrates the overall approach.

6. Conclusion

We presented the ZeroPlane Planar Benchmark and the ZeroPlane framework for in-the-wild single-image 3D plane reconstruction. The benchmark provides $\sim$560K dense plane annotations at 640×480 across 14+ datasets, enabling the first large-scale cross-domain training and zero-shot evaluation for plane reconstruction. ZeroPlane couples DINOv2 features with decoupled exemplar-guided normal/offset prediction and geometry-enhanced plane queries, achieving consistent state-of-the-art zero-shot performance across all four evaluation benchmarks. Ablations confirm the contribution of each component and the importance of large-scale mixed-domain training and high-resolution inputs. Future work includes extending annotations to additional modalities such as LiDAR, exploring semi-supervised labeling to scale further, and investigating temporal consistency for video-based plane reconstruction. We will publicly release the ZeroPlane benchmark and pre-trained models to facilitate further research in cross-domain 3D scene understanding.

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Table 2. Zero-shot evaluation of different methods or settings on indoor datasets (NYUv2 [27], 7-Scenes [26]) and outdoor datasets (ParallelDomain [13, 21], ApolloScape [16]). (S: trained on ScanNetv1 [9]; S-v2: trained on ScanNetv2, whose training set is much larger than ScanNetv1; M: trained on mixed datasets.)

Evaluation Dataset (Indoor)	Method	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI (↑)	VOI (↓)	SC (↑)	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
NYUv2 [27]	PlaneRCNN (S-v2) [19]	0.84	1.60	0.61	3.04	9.82	**43.41**	9.24	24.12	43.60
	PlaneAE (S) [37]	0.89	1.40	0.68	1.26	3.36	15.06	11.18	26.54	38.52
	PlaneTR (S) [30]	0.89	1.16	0.71	2.14	6.33	28.30	12.42	27.01	34.63
	PlaneRecTR (S) [25]	**0.92**	1.05	**0.75**	3.3	10.13	41.47	16.42	34.42	45.7
	Ours-DINO-B (S)	**0.92**	**1.02**	**0.75**	**4.33**	**11.03**	41.45	**24.05**	**39.88**	**47.35**
	PlaneRecTR (M) [25]	0.91	1.01	0.73	5.77	14.29	52.01	24.97	40.96	54.71
	Ours-DINO-B (M)	**0.92**	**0.93**	**0.75**	8.54	17.86	55.08	37.29	47.58	57.19
	Ours-DINO-L (M)	**0.92**	0.96	**0.75**	9.32	20.29	55.37	38.15	**47.76**	**56.3**
	Ours-Dust3R (M)	**0.92**	0.97	**0.75**	**9.94**	**21.2**	**56.03**	**38.32**	47.6	56.14
7-Scenes [26]	PlaneRecTR (M) [25]	**0.92**	**1.04**	0.78	3.20	10.97	45.30	11.17	26.67	42.17
	Ours-DINO-B (M)	**0.92**	1.06	**0.79**	8.19	17.19	48.13	23.96	35.43	44.17
	Ours-DINO-L (M)	**0.92**	**1.04**	0.78	7.16	16.25	48.74	25.51	**36.32**	**44.49**
	Ours-Dust3R (M)	**0.92**	1.08	0.78	**9.11**	**19.14**	**49.73**	**26.69**	36.11	44.43
Evaluation Dataset (Outdoor)	**Method**	**Plane Segmentation**			**Plane Recall (depth)**			**Plane Recall (normal)**		
		RI (↑)	VOI (↓)	SC (↑)	@1m	@3m	@10m	@5°	@10°	@30°
ParallelDomain [13, 21]	PlaneRecTR (M) [25]	0.92	0.96	0.76	19.11	35.49	51.92	24.17	35.11	53.36
	Ours-DINO-B (M)	0.94	0.65	0.85	25.96	44.51	63.45	53.23	65.74	69.62
	Ours-DINO-L (M)	**0.95**	**0.57**	**0.87**	**27.66**	44.72	**64.0**	52.47	**66.13**	**70.47**
	Ours-Dust3R (M)	0.94	0.69	0.84	27.4	**46.26**	63.19	**53.91**	65.36	69.87
ApolloScape [16]	PlaneRecTR (M) [25]	0.94	0.38	0.91	10.09	43.06	57.06	22.99	33.32	51.88
	Ours-DINO-B (M)	**0.96**	**0.3**	**0.93**	10.28	40.44	57.34	27.56	39.78	56.05
	Ours-DINO-L (M)	**0.96**	**0.3**	**0.93**	**12.26**	**46.4**	**62.08**	**28.51**	**40.32**	**59.87**
	Ours-Dust3R (M)	0.95	0.35	0.92	10.96	44.21	58.9	24.43	37.3	55.15

Table 3. Ablation studies on the contributed components under both single-dataset (ScanNet) training and mixed-dataset training schemes, evaluated on the NYUv2 dataset.

Training datasets	Component			Plane Recall (depth)			Plane Recall (normal)		
	DPT	Cls-Reg	Geo-Attn	@0.05m	@0.1m	@0.6m	@5°	@10°	@30°
ScanNet	-	-	-	2.76	8.21	39.37	10.87	29.61	45.1
	✓	-	-	4.17	10.66	41.2	19.78	37.93	47.0
	✓	✓	-	3.88	9.71	41.27	**24.27**	38.73	45.84
	✓	✓	✓	**4.33**	**11.03**	**41.45**	24.05	**39.88**	**47.35**
Mixed datasets	-	-	-	3.9	11.65	48.65	20.54	38.15	53.1
	✓	-	-	6.1	15.65	54.18	29.55	45.23	56.59
	✓	✓	-	7.61	16.93	53.97	36.11	46.36	55.39
	✓	✓	✓	**8.54**	**17.86**	**55.08**	**37.29**	**47.58**	**57.19**

Table 4. Ablation study on training our framework using different combinations of indoor and outdoor datasets.

Evaluation Dataset	Training Datasets	Plane Recall (depth)			Plane Recall (normal)		
		@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2 [27]	Indoor Only	8.41	**18.7**	**55.89**	**37.82**	**47.93**	57.08
	Outdoor Only	0.43	1.71	15.59	7.9	14.68	26.75
	Starter Mixed (110K)	5.24	13.43	46.26	31.2	43.14	51.04
	Full Mixed (560K)	**8.54**	17.86	55.08	37.29	47.58	**57.19**
ParallelDomain [13, 21]	Outdoor Only	**26.72**	43.23	59.87	48.55	59.32	64.43
	Indoor Only	0.81	1.19	3.66	16.94	21.4	37.57
	Starter Mixed (110K)	14.04	35.45	53.53	42.54	55.91	61.36
	Full Mixed (560K)	25.96	**44.51**	**63.45**	**53.23**	**65.74**	**69.62**

Table 5. Comparison on training using low vs. high resolution data, then evaluation on high-resolution groundtruth.

Evaluation Dataset	Training Resolution	Plane Segmentation			Plane Recall (depth)			Plane Recall (normal)		
		RI (↑)	VOI (↓)	SC (↑)	@0.05m / 1m	@0.1m / 3m	@0.6m / 10m	@5°	@10°	@30°
NYUv2	Low-res	**0.92**	0.93	0.76	**8.35**	**17.78**	55.33	37.41	47.87	57.35
	High-res	**0.92**	**0.84**	**0.77**	7.47	16.19	**57.31**	**40.09**	**50.34**	**59.87**
ParallelDomain	Low-res	**0.94**	0.68	0.84	26.38	44.43	62.68	51.62	64.34	68.43
	High-res	**0.94**	**0.63**	**0.85**	**30.38**	**50.0**	**66.09**	**56.55**	**67.91**	**71.32**

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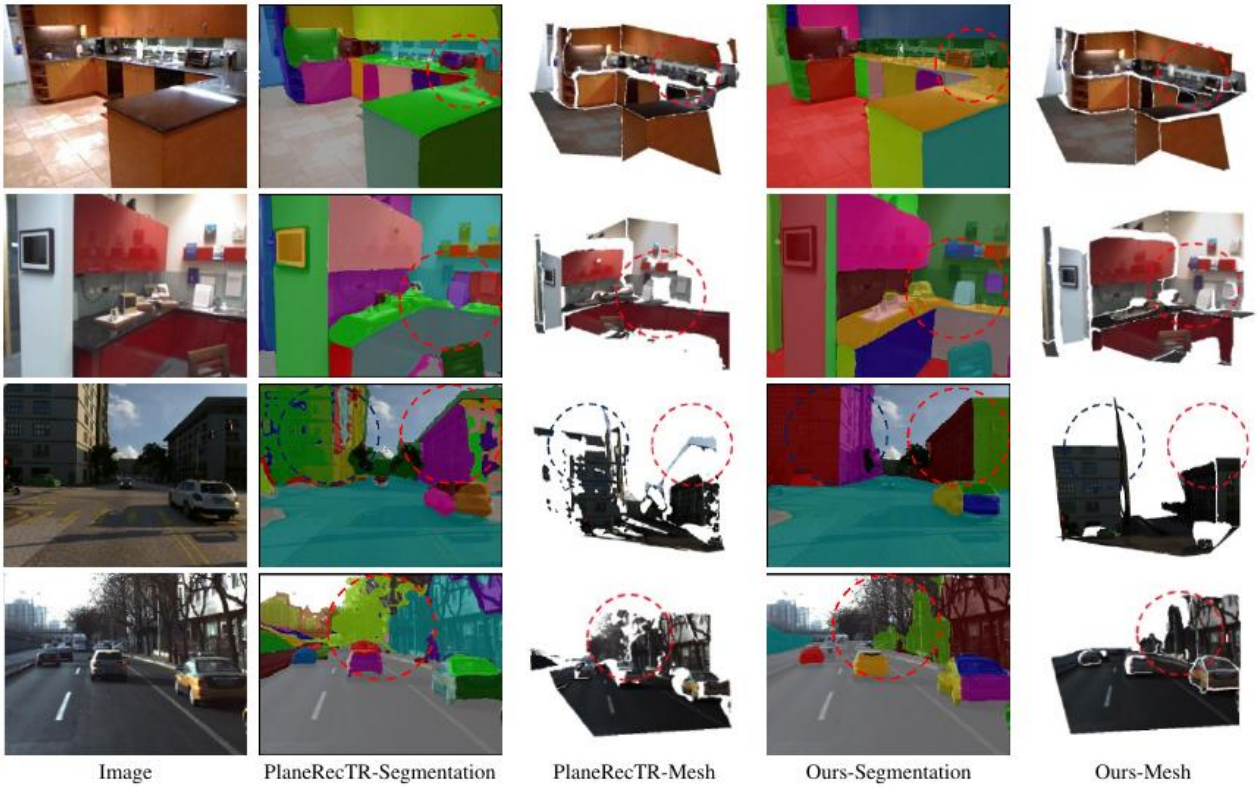

Figure 2. Qualitative zero-shot results on NYUv2, 7-Scenes, ParallelDomain, and ApolloScape (top to bottom). ZeroPlane produces accurate plane segmentation and mesh reconstruction across diverse indoor and outdoor scenes without having seen these domains at training time.

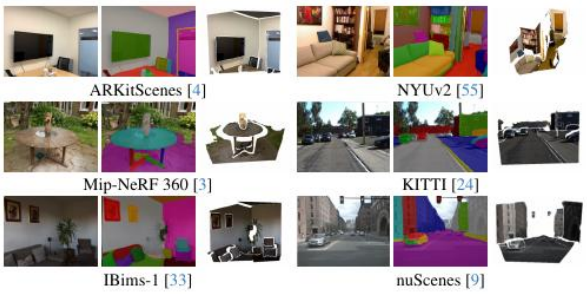

Figure 3. ZeroPlane demonstrates zero-shot generalizability on unseen indoor and outdoor environments. Mixed-domain training with geometry-enhanced queries enables robust in-the-wild plane reconstruction.

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